

Figure 6: *Make* command console output screen

Figure 7: Sample *helloworld* module insert/remove

Figure 8: Sample *helloworld* module with passing runtime argument

- For an example of how to use module management commands, please refer to Figure 3.

Let's write a simple kernel module and see how easy and interesting it is. Refer to Figure 5, where the simple module is written.

- *hello_init()*: This is called when the module is inserted into the kernel using *insmod*. This function gets invoked by the 'module_init' macro. The *init* function is responsible for registering the module with the kernel.
- *hello_exit()*: This function is called when the module

- *Macros module_init(hello_init) & Module_init(hello_exit)*: Using these macros, programmers can give user defined names to the *init* and *cleanup* functions. These macros are defined in `<linux/init.h>`.
- *Printk*: In kernel module programming, 'printk' is used to print kernel messages in to the kernel logs. *Printk* messages are linked to the priority associated with them. For all behavioural purposes, we use 'printk' in kernel module programming much as we use 'printf' in user level C programs.

Once the module is compiled and built using *make*, the *'module.ko (helloworld.ko)'* will be created.

Now that we have the *helloworld.ko* file, insert this module into or remove it from the kernel by using the *insmod/rmmod* commands. Please refer the Figure 7 for this.

As with any other program, run time arguments can be passed to the kernel module also, as follows:

Figure 8 shows the sample output of passing runtime arguments for the *helloworld* kernel module. **END** 

The Linux Kernel Module Programming Guide: <http://www.tldp.org/LDP/kmpg/2.6/kmpg.pdf>

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